Spoken Tutorial – Introduction to Retrieval-Augmented Generation (RAG)

Title: Introduction to Retrieval-Augmented Generation (RAG)

Module	Artificial Intelligence Concepts
Episode	Retrieval-Augmented Generation (RAG)
Learning Objective	At the end of this tutorial learner will be able to 1. Understand the core concept and benefits of Retrieval-Augmented Generation (RAG).
Approx. Duration	15-20 minutes
Outline	• The Challenge of 'Closed-Book' AI • Introducing RAG: An Open-Book Approach • How RAG Enhances LLMs • The 'Retrieval' Component: The Librarian • What the Retriever Does • The 'Generation' Component: The Eloquent Expert • LLM's Role in Generation • The 3-Step RAG Flow • Benefits of RAG: Truthful and Specific • RAG: Foundation for 'Chat with your Data'
Meta Tags	RAG, Retrieval-Augmented Generation, LLM, Large Language Models, AI, Artificial Intelligence, Knowledge Retrieval, Generative AI, Information Retrieval, Spoken Tutorial
Pre-requisite Tutorial	Basic understanding of Large Language Models (LLMs).

Script

Visual Cue	Narration
	Welcome to this **Spoken Tutorial** on **Retrieval-Augmented Generation**, also called **RAG**. In this tutorial, we will learn how **AI** models can give more correct and helpful answers.
	After this tutorial, you will understand what **Retrieval-Augmented Generation**, or **RAG**, is. You will also know its main benefits.
	You do not need to install any special software for this tutorial. You only need a web browser. You can use it to find more information about **AI** and **LLMs** if you want.

	To follow this tutorial, you should know what **Large Language Models**, or **LLMs**, are. You should also know how they generally work.
An illustration of a confused person looking at a computer screen displaying an incorrect or outdated AI response.	Have you ever asked an **AI** a question and got a wrong or old answer? This happens when **AI** models use only the knowledge they learned before.
An illustration of a student confidently answering an exam question while looking at an open textbook beside them.	**Retrieval-Augmented Generation**, or **RAG**, helps solve this 'closed-book' problem. Imagine it like an open-book exam for an **AI**. The **AI** first looks for information, instead of just using its stored knowledge.
An illustration showing an AI brain icon connected to a stack of documents, with an arrow pointing to a speech bubble representing an answer.	**RAG** connects outside knowledge with **AI** models. It makes **LLMs** work better by giving them real documents. The **AI** gets these documents before it creates an answer.
An illustration of a friendly librarian searching through rows of books in a vast library, with a magnifying glass icon.	The first part of **RAG** is called 'Retrieval'. This part works like a careful librarian. It searches a set of documents, like **PDFs** or company wikis. It does this to find answers to your question.
An illustration of a hand pulling a specific document from a stack of many documents, with a search icon nearby.	When you ask a question, the retriever quickly finds exact paragraphs or documents. These documents may have the answer. It is like finding the correct page in a book.
An illustration of a person speaking eloquently, surrounded by thought bubbles containing facts and documents.	The second part is called 'Generation'. Here, the **Large Language Model**, or **LLM**, takes over. It acts like a smart expert who has just received all the facts.
An illustration of an LLM icon (like a robot head) receiving a question bubble and several document icons, then generating an answer bubble.	The **LLM** gets your question and the documents found by the librarian. Its main job is to answer your question only from these documents. It does not use its own stored knowledge for this.
A three-step infographic showing 'Query' -> 'Retrieve' -> 'Generate' with arrows connecting them.	The **RAG** process has three simple steps. First, you give a question, called a Query. Second, the system finds information, this is Retrieve. Third, the system creates an answer using that information, this is Generate.
An illustration of a checkmark icon surrounded by documents and a speech bubble with a correct answer, symbolizing truth and accuracy.	**RAG** makes **AI** answers more correct, clear, and provable. It uses outside information, so the **AI** gives exact answers that fit the topic. It can even tell you which documents it used.
An illustration of a chat interface overlaid on a stack of diverse data documents (PDFs, spreadsheets, web pages).	This method is the base for many 'Chat with your data' tools. Imagine asking questions to all your company's documents. You will get exact answers with their sources.

	In this tutorial, we learned about **Retrieval-Augmented Generation**, or **RAG**. We saw how **RAG** uses found information with large language models. This helps **AI** give correct, provable, and helpful answers. It solves the problem of **AI** using only its old knowledge.
	Think of a place where **RAG** would be very useful. For example, how can it make customer support better for a product? Write down your ideas.
	This **Spoken Tutorial** is from **EduPyramids Educational Services Private Limited**. It is from **SINE**, **IIT Bombay**. Thank you for joining us!